Physics of the teeterboard

Contact, rotation and work in a loaded descent

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Abstract

We derive the mechanism behind part 1 of Mad Scientist 1, from an illustrative human teeterboard launch to its computed two-weight lever. Independent integrations reproduce 6.994139 J of receiver work and event COP 1.188267. Contact, rebound and renewed contact explain the transfer. The drivers' further descent and final stores resolve the work account. Figures and tables connect the force laws, motion and measurable signals.

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Latest on GitHub: https://github.com/hobnilre/physics-teeterboard (planned; local edition)

1 From a launch to a work question

Two people land on one side of a teeterboard and lift a third. The pivot redirects motion while legs and board deform. Part 1 of Mad Scientist 1 uses this picture to motivate a falling-weight lever [OP-CT-HW-01]. Its human speeds are illustrative; its numerical study concerns a separate mechanical model.

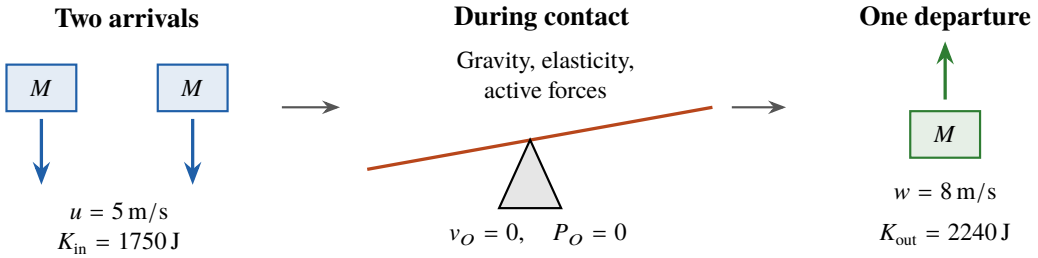


Figure 1. Schematic of the illustrative launch. Incoming and outgoing kinetic energies describe different endpoints; compression, gravity and any muscular work act between them.

For three equal masses $M = 70\,\mathrm{kg}$, two arrivals at downward speed $u = 5\,\mathrm{m/s}$ and one departure at upward speed $w = 8\,\mathrm{m/s}$ give

$$K_{\mathrm{in}} = Mu^2 = 1750\,\mathrm{J}, \qquad K_{\mathrm{out}} = \tfrac{1}{2}Mw^2 = 2240\,\mathrm{J}, \quad \frac{K_{\mathrm{out}}}{K_{\mathrm{in}}} = \frac{32}{25}. \tag{1}$$

The difference is 490 J. Two such masses descending another 0.50 m release $2Mg(0.50\,\mathrm{m}) = 686.7$ J. This illustrates the scale of gravity's contribution; the flyer's rise, remaining motion, deformation, losses and active work must also enter a complete launch account.

2 Geometry and forces of the computed event

The calculation replaces the performers with two identical, vertically guided drivers and an external viscous receiver. The rigid board has a fixed pivot, inertia I, a torsion spring K_ϕ and bearing damping c_b. Its centre of mass lies at the pivot. Contacts exert vertical forces; ideal guides do no work. Board bending, a departing flyer and active leg forces are absent from this model.

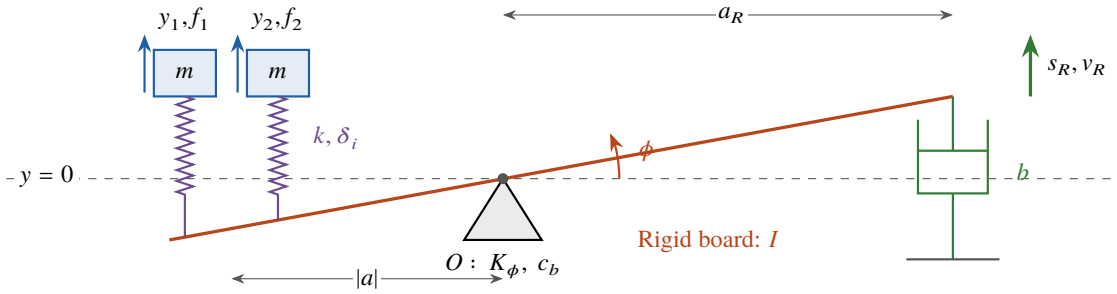


Figure 2. Model schematic, shown in contact. The two drivers share the left arm but have separate vertical springs. Positive rotation lowers that arm and raises the receiver. The pivot spring and bearing act relative to ground.

Upward driver heights are y_i and positive board angle is ϕ. A signed arm a has height $a \sin\phi$; the receiver has displacement $s_R = a_R \sin\phi$. Compression and receiver velocity follow from geometry:

$$\delta_i = [a \sin\phi - y_i]_+, \quad f_i = k\delta_i, \qquad v_R = a_R \cos\phi\, \dot\phi, \quad F_R = b v_R. \tag{2}$$

Here $[x]_+ = \max(x, 0)$ and F_R is the force delivered to the receiver. The opposite force acts on the board. Multiplying each vertical force by its horizontal moment arm gives Newton's equations:

$$\begin{aligned} m\ddot y_i &= -mg + f_i, \qquad i = 1, 2, \\ I\ddot\phi &= -a\cos\phi\,(f_1 + f_2) - K_\phi\phi - c_b\dot\phi - b a_R^2 \cos^2\phi\,\dot\phi. \end{aligned} \tag{3}$$

Table 1. Reference parameters. Initially $y_1 = y_2 = H$, $\phi = 0$ and all velocities are zero. The event ends at the fixed time $T = 1.5$ s.

Quantity	Value	Quantity	Value
Each driver m	1 kg	Board inertia I	0.15 kg m^2
Driver arm a	−0.50 m	Receiver arm a_R	0.80 m
Contact stiffness k	500 N/m	Load b	32 N s/m
Torsion K_ϕ	50 N m/rad	Bearing c_b	0.10 N m s/rad
Release height H	0.30 m	Gravity g	9.81 m/s^2

The springs push only while compressed. At separation $\delta_i = f_i = 0$; the law introduces neither an impact impulse nor a discrete energy loss. Both contacts are purely elastic. Dissipation occurs in the receiver and bearing.

3 The contact history

Before touchdown the board remains horizontal and each driver is in free fall: $y = H - gt^2/2$. Contact begins at $t_c = \sqrt{2H/g} = 0.247310$ s, with $\dot{y}(t_c) = -\sqrt{2gH} = -2.426108$ m/s. The two contacts are simultaneous by symmetry. Their forces decelerate the drivers and apply a positive torque to the board; the right arm rises against the receiver.

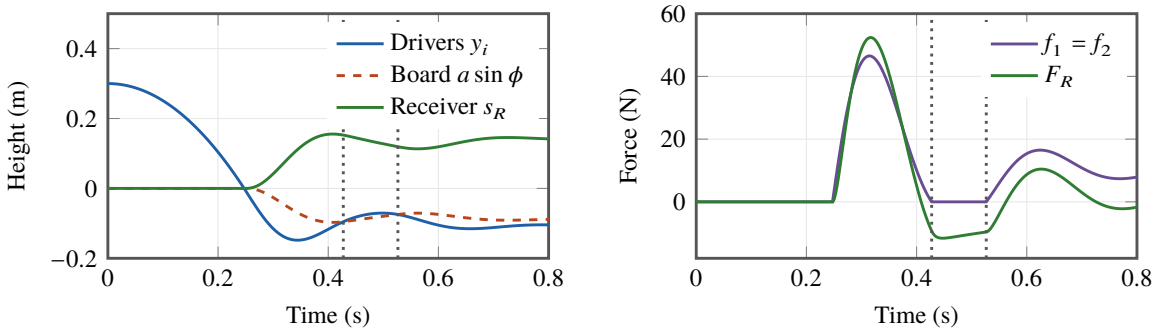


Figure 3. Computed reference trajectory. Left: driver, left board contact and receiver heights. Right: upward force on each driver and signed force delivered to the receiver. The contact-free interval lies between the dotted lines.

Table 2. Numerically located transitions and the prescribed endpoint. Driver and board reversals occur at different times.

Event	Time (s)	Driver y (m)	Board ϕ (rad)
First contact	0.247310	0.000000	0.000000
First driver reversal	0.344255	−0.148164	0.139293
First board reversal	0.408321	−0.110027	0.195639
Contact opens	0.427352	−0.095356	0.191887
Contact closes again	0.526451	−0.074597	0.149754
Declared event end	1.500000	−0.114406	0.190618

The drivers reach their lowest position while the board is still rotating upward on the receiver side. Elastic unloading then lifts the drivers. They leave the board, follow a short ballistic arc, and contact it again. Smaller oscillations follow. The peak force on each driver is 46.53 N and the largest compression is 93.05 mm; these are spring-model predictions.

Receiver force changes sign when receiver velocity changes sign, but its power remains $P_R = F_R v_R = bv_R^2 \geq 0$. The descending return motion therefore also delivers work. A one-way receiver or a free flyer would require a different law.

Moving board points have $v_a = a \cos\phi\dot{\phi}$ and $|v_R/v_a| = a_R/|a| = 1.6$. Here the drivers can compress and leave the contact, so their speed is not the board-point speed. Compliance and timing determine the transfer; the arm ratio alone cannot predict the outgoing motion or work.

A stationary ideal pivot supplies impulse $J_O = \int F_O\, dt$ but no work, since $P_O = F_O v_O = 0$. The performers' vertical momentum alone is not conserved. Outgoing speed does not identify the work source; the contact trajectory reveals the forces and displacements.

4 Deriving the work account

The mechanical boundary contains the drivers, board, contact and torsion springs, with gravity included as potential energy. The receiver and bearing heat lie outside. Set gravitational zero at the initial horizontal board:

$$U = \tfrac{1}{2}m\sum_i \dot{y}_i^2 + \tfrac{1}{2}I\dot{\phi}^2 + mg\sum_i y_i + \tfrac{1}{2}K_\phi\phi^2 + \tfrac{1}{2}k\sum_i \delta_i^2. \tag{4}$$

During contact, $\dot{\delta}_i = a\cos\phi\,\dot{\phi} - \dot{y}_i$. Multiply (3) by the corresponding velocities and differentiate (4): the contact and torsion powers cancel internally. Hence

$$\dot{U} = -bv_R^2 - c_b\dot{\phi}^2, \qquad U(T) + W_R + Q_b = U(0), \quad (W_R, Q_b) = \int_0^T (bv_R^2, c_b\dot{\phi}^2)\,dt. \tag{5}$$

The same identity holds across opening and closure because spring energy vanishes at zero compression. This balance follows from the force laws.

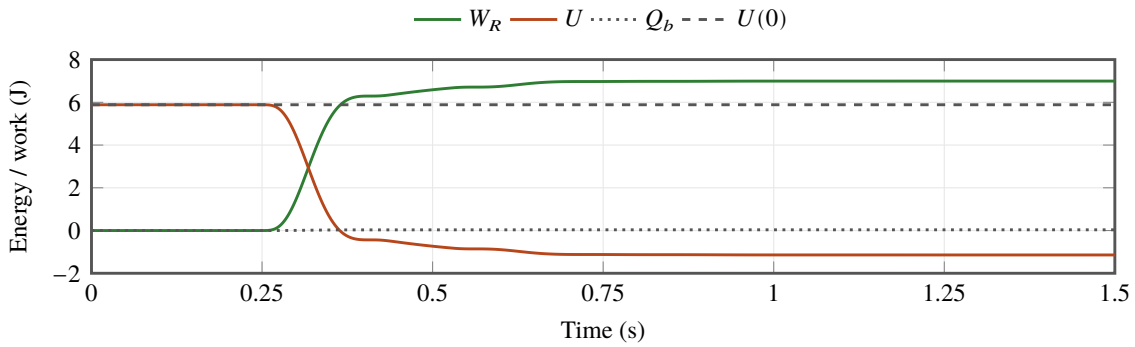


Figure 4. Computed cumulative receiver work and total mechanical storage. The latter includes signed gravitational potential. Their sum, plus bearing heat, stays at 5.886 J.

Table 3. Endpoint stores evaluated directly from the solved state; work obtained by power integration.

Energy or work	Start (J)	End or accumulated (J)
Gravitational potential	5.886000	−2.244651
Kinetic energy, both drivers and board	0	0.000012
Torsion spring	0	0.908383
Contact springs, both	0	0.193519
Total mechanical storage U	5.886000	−1.142737
Receiver work W_R; bearing heat Q_b	0; 0	6.994139; 0.034598

The extra 2.244651 J of gravitational descent below the ready plane pays for 1.108139 J of output above the starting reference, 1.101914 J of remaining kinetic and elastic energy, and 0.034598 J of bearing heat. The mechanism has a resolved work account, including its negative final potential.

5 What the ratio and the load change

There is no external actuator during this event. Its declared starting-energy denominator is $D = 2mgH = 5.886$ J. Retaining exactly that reference gives

$$C_{\text{event}} = \frac{W_R}{D} = 1.188267, \qquad \eta_g = \frac{W_R}{mg\sum_i [y_i(0) - y_i(T)]} = 0.860219. \tag{6}$$

These answer different questions: output relative to the initial ready-plane reference, and the fraction of the actual 8.130651 J gravitational decrease delivered to the receiver. A shifted datum $y' = y - h$ changes D to $D' = D - 2mgh$ and $U(T)$ by the same constant. Work and η_g stay unchanged; W_R/D' depends on the reference and requires $D' > 0$.

Table 4. Reproduction of the reference case and all four confirmation cases in the companion study; the event and height reference are fixed.

k (N/m)	b (N s/m)	Receiver work (J)	Event COP
500	32	6.994139	1.188267
450	30	7.013120	1.191492
450	34	7.017282	1.192199
550	30	6.974505	1.184931
550	34	6.978687	1.185642

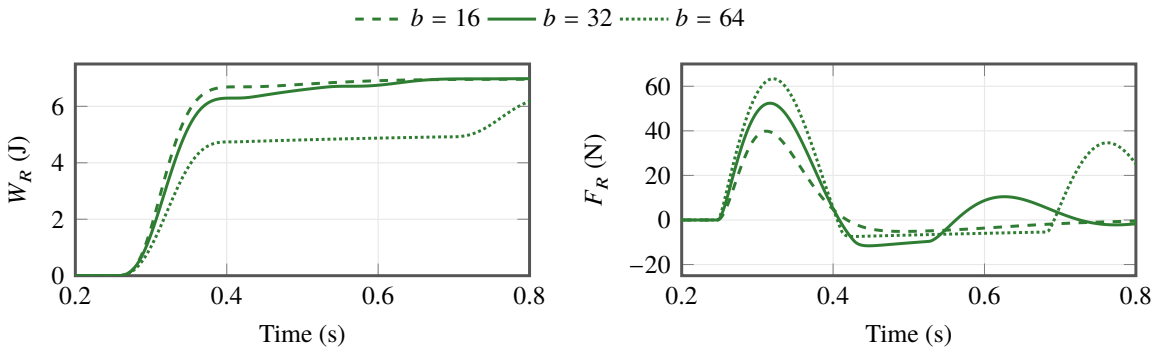


Figure 5. Computed load controls at $k = 500$ N/m. Higher resistance changes the force pulse and delays work transfer; a larger force need not yield more work by 1.5 s.

At $b = 16, 32, 64$ N s/m the work is respectively 6.959211, 6.994139 and 6.977427 J. For $b = 0$, receiver work is zero and substantial motion remains. With a stationary loaded equilibrium, $f_i = mg$ gives $K_\phi \phi_* = -2amg\cos\phi_*$ and $y_* = a\sin\phi_* - mg/k$. The reference values are $\phi_* = 0.192573$ rad and $y_* = -0.115313$ m, close to, but distinct from, the finite-time endpoint.

We integrated the symmetric Newton equations independently with two numerical methods and timestep refinement: 21 runs covering all eight configurations. The largest receiver-work spread was 2.66×10^{-7} J; maximum storage residual was 3.22×10^{-7} J. Independent power quadrature differed by at most 1.05×10^{-11} J. These estimate numerical sensitivity; they do not bound errors in a physical contact law.

6 Measurements and the next event

A synchronized load cell and displacement encoder give signed receiver work, $W_R = \int F_R \, ds_R$, including the return stroke. Driver heights and board angle distinguish gravity's contribution from compression and remaining motion; an endpoint speed alone cannot.

Table 5. Predicted signal scales and the companion's prospective instrument budget. The bounds require calibration; they are not measurements made here.

Signal or condition	Reference prediction or assumed requirement
Peak receiver force and speed	52.40 N; 1.637 m/s
Receiver displacement range; total travel	0–0.1555 m; 0.2460 m
Contact and event timing	Separation 0.427–0.526 s; end 1.5 s
Prospective acquisition	200 Hz response; 2 kHz sampling; skew ≤ 0.1 ms
Assumed simultaneous energy bounds	$B_{\text{in}} = 0.02943$ J; $B_{\text{out}} = 0.06608$ J

That budget includes initial mass and height, force and displacement gains, offset, sampling, timing, bandwidth, loading and event-boundary allowances. With those simultaneous bounds the reference result remains above 1.15:

$$\frac{W_R - B_{\text{out}}}{D + B_{\text{in}}} = 1.17118 > 1.15, \qquad W_R - 1.15D = 0.225239\,\text{J}. \tag{7}$$

Measuring the drivers' final height is equally decisive: the predicted mean is 0.114406 m below the ready plane. It locates the energy omitted when the initial height reference is mistaken for the total descent.

6.1 Preparation and reset

At T the board is tilted and the contacts remain compressed. Another identical release needs a reset. The companion's separately prescribed preparation and reset calculation supplies the account below.

Table 6. Retained full-history calculation in the companion study. Total bearing heat is 0.039844 J and returned actuator work is 0.000012 J; initial and final stores are zero.

Interval	Gross supplied work (J)	Receiver work (J)
Preparation to the stated initial state	5.886000	0
The 1.5 s event	0	6.994139
Prescribed reset	2.210180	1.062185
Full history	8.096180	8.056324

The full-history ratio is $8.056324/8.096180 = 0.995077$: reset adds both input and output, with different endpoint states from (6). Human launches also involve leg activation, board bending and flyer takeoff. In this model, contact elasticity times the transfer, the lever redirects it, and the receiver extracts work from a descent extending below the starting reference.

Reference

1. Hob Nilre and Bo C. Herlin (2026), *Mad Scientist 1*, part 1, "The falling-weight lever," and its common simulation account.